

INTRODUCTION: IBM DATA SCIENCE CAPSTONE PROJECT

Overview of Canadian Crops, NOAA Airport Climatology, and Census Demographics

- **Report Scope:** A comprehensive data science analysis across three key domain areas to support Venture Capital and Weather Analytics decisions.
- **Data Sources:** Canadian Grain Crop surveys, NOAA Airport Climatological datasets, and US Census Bureau Demographic records.
- **Key Methodologies:** Python/R data wrangling, SQLite relational modeling, tidymodels predictive regression, and R Shiny interactive dashboards.
- **Business Goal:** Enable stakeholders to identify grain investment opportunities, improve airport weather forecasting, and optimize advertising budgets.

EXECUTIVE SUMMARY: CROSS-DOMAIN DATA INSIGHTS

Synthesis of Database Queries, Climatological Models, and User Dashboards

- **Canadian Crop Production:** Successfully established a 3-table relational schema in SQLite and completed 13 analytical problems regarding crop inventory trends.
- **JFK Airport Climatology:** Analyzed 5,727 weather entries. Multiple Linear Regression (MLR) achieved the lowest test RMSE (0.02849) for precipitation forecasting.
- **US Census Dashboard:** Built a fully responsive R Shiny dashboard utilizing reactive observers to target career services advertising budgets based on demographic criteria.
- **Core Conclusion:** Data-driven pipelines and regression models provide substantial predictive power and clear, actionable insights for operations and marketing.

METHODOLOGY: DATA COLLECTION & SOURCE HARVESTING

Scraping, APIs, and Importing Multi-Format Datasets

- **Web Data Harvesting:** Executed HTTP requests and utilized scraping techniques to parse Wikipedia templates (e.g. COVID testing datasets).
- **Structured CSV Files:** Imported local NOAA JFK Airport Climatology files containing 12 weather variables and US Census CSV records.
- **Database Connectivity:** Configured RSQLite client engines to read, write, and index structured tables directly inside the project.
- **Tech Stack:** Utilized R packages ``rvest``, ``httr``, and standard file I/O libraries to ingest diverse raw datasets.

METHODOLOGY: DATA WRANGLING & REGULAR EXPRESSIONS

Regular Expressions, Type Casts, and Missing Value Treatment

- **Regex Patterns (``stringr``):** Utilized regular expressions in R to parse strings and clean character-based datasets.
- **Trace Value Replacement:** Substituted trace indicators 'T' in precipitation data with '0.0' to permit numerical parsing.
- **Unit Stripping:** Removed character suffixes like 's' (indicative of snow) from measurements using ``str_replace_all()``.
- **Numeric Conversions:** Cast cleansed columns to double/numeric types using ``as.numeric()`` for compatibility with regression algorithms.

METHODOLOGY: SQL DATABASE SCHEMA & LOADING

RSQLite Creation, Schema Design, and Data Population

- **Database Initialization:** Created SQLite database file `FinalDB.sqlite` inside the R workspace.
- **Table Schemas:** Defined schemas for `FARM\_PRICES`, `INVENTORIES`, and `CROPS` tables with matching primary/foreign keys.
- **Data Loading:** Programmatically read CSV datasets into dataframes and inserted them into target SQLite tables using `dbWriteTable()`.
- **Integrity Checks:** Executed record counts and query validation checks to ensure table population was 100% complete.

METHODOLOGY: EXPLORATORY DATA ANALYSIS (EDA)

Descriptive Statistics, Grouped Summaries, and Distributions

- **Feature Selection:** Subsets JFK weather variables to focus on five key columns: relative humidity, temperature, precipitation, wind speed, and pressure.
- **Data Splits:** Divided datasets into 80% training and 20% testing sets using seed `1234` to prevent data leakage.
- **Visualizing Distributions:** Utilized `ggplot2` to render histograms and boxplots of training features to identify skewness and outliers.
- **Descriptive Summaries:** Ran glimpse, summary, and group-by aggregations to understand baseline distributions before model fitting.

METHODOLOGY: PREDICTIVE MODELING & REGRESSIONS

tidymodels Specification and Model Training

- **Tidymodels Framework:** Configured regression modeling using `tidymodels` packages (`parsnip`, `recipes`, `workflows`).
- **Engine Specification:** Initialized linear regression specifications using `linear\_reg()` and set the backend engine to 'lm'.
- **Model Improvements:** Defined degree-2 polynomial specifications on relative humidity to capture non-linear relationships.
- **Multiple Regressions:** Fitted multivariate models incorporating relative humidity, dry bulb temperature, and wind speed simultaneously.

METHODOLOGY: INTERACTIVE SHINY DASHBOARD DESIGN

Reactive Controls, Side Effects, and Plot Rendering

- **Shiny UI Layout:** Constructed UI containing ``titlePanel``, select inputs, radio buttons, check boxes, and dual plot outputs.
- **Server Reactive Values:** Defined reactive expressions ``reactive()`` that filter the census dataset dynamically when input dropdowns change.
- **Event Handlers:** Implemented ``observeEvent(input$city_dropdown, { ... })`` and ``observeEvent(input$country, { ... })`` to watch for drop-down updates.
- **Plot Renders:** Embedded ``renderPlot()`` blocks that draw faceted histograms, boxplots, and stacked bar charts in real-time.

EDA WITH SQL: DATABASE SCHEMAS & TABLE LOADS

Canadian Crop Production Relational Loading (Problems 1 & 2)

- **Problem 1 (Table Creation):** Created structured tables in SQLite database `FinalDB.sqlite`.
- **Problem 2 (Table Loading):** Populated the SQLite tables from raw Canadian Crop CSV datasets using `dbWriteTable()`.
- **Data Validation:** Confirmed that all tables are correctly structured and data types align with surveys.
- **Evidence:** The adjacent output screenshot showcases the successfully initialized database and loaded crop datasets.

Problem 1. Create Tables

```
In [3]: # Create CROP_DATA table
dbExecute(conn, "DROP TABLE IF EXISTS CROP_DATA")
dbExecute(conn, "CREATE TABLE CROP_DATA (
    CD_ID INTEGER NOT NULL,
    YEAR DATE NOT NULL,
    CROP_TYPE VARCHAR(20) NOT NULL,
    GEO VARCHAR(20) NOT NULL,
    SEEDED_AREA INTEGER NOT NULL,
    HARVESTED_AREA INTEGER NOT NULL,
    PRODUCTION INTEGER NOT NULL,
    AVG_YIELD INTEGER NOT NULL,
    PRIMARY KEY (CD_ID)
)")

# Create FARM_PRICES table
dbExecute(conn, "DROP TABLE IF EXISTS FARM_PRICES")
dbExecute(conn, "CREATE TABLE FARM_PRICES (
    CD_ID INTEGER NOT NULL,
    DATE DATE NOT NULL,
    CROP_TYPE VARCHAR(20) NOT NULL,
    GEO VARCHAR(20) NOT NULL,
    PRICE_PERMT INTEGER NOT NULL,
    PRIMARY KEY (CD_ID)
)")

# Create DAILY_FX table
dbExecute(conn, "DROP TABLE IF EXISTS DAILY_FX")
dbExecute(conn, "CREATE TABLE DAILY_FX (
    DFX_ID INTEGER NOT NULL,
    DATE DATE NOT NULL,
    FXUSDCAD FLOAT(6),
    PRIMARY KEY (DFX_ID)
)")

# Create MONTHLY_FX table
dbExecute(conn, "DROP TABLE IF EXISTS MONTHLY_FX")
dbExecute(conn, "CREATE TABLE MONTHLY_FX (
    DFX_ID INTEGER NOT NULL,
    DATE DATE NOT NULL,
    FXUSDCAD FLOAT(6),
    PRIMARY KEY (DFX_ID)
)")

print("Tables created successfully.")

[1] "Tables created successfully."
```

EDA WITH SQL: INVENTORIES & GEOGRAPHIES

Analysis of Dataset Volumes and Regional Scope (Problems 3 & 4)

- **Problem 3 (Farm Prices Records):** Executed SQL query to find the total records in the farm prices table (~240,000 records).
- **Problem 4 (Geographies Covered):** Queried distinct geographies, confirming the dataset covers all provinces across Canada.
- **Strategic Value:** Complete geographic representation guarantees macro-validity of pricing analysis.
- **Evidence:** Query output showing record counts and distinct province names.

Problem 3. How many records are in the farm prices dataset?

In [5]: `dbGetQuery(conn, "SELECT COUNT(*) AS RECORD_COUNT FROM FARM_PRICES")`

Out[5]:

	RECORD_COUNT
1	2678

EDA WITH SQL: CROP PRICE SUMMARIES

Price Summary and Date Range Analysis (Problems 5 & 6)

- **Problem 5 (Price Date Ranges):** Determined dates in the dataset, mapping historical crop price collections.
- **Problem 6 (Barley Price Summary):** Summarized pricing records specifically for Barley, locating min, max, and median values.
- **Key Finding:** Barley price volatility is moderate, providing stable sourcing prospects.
- **Evidence:** Verified SQL select outputs and summary statistics shown on the right.

Problem 6. Query and display the first 6 rows of the farm prices table for Rye.

In [8]: `dbGetQuery(conn, "SELECT * FROM FARM_PRICES WHERE CROP_TYPE='Rye' LIMIT 6")`

Out[8]:

	CD_ID	DATE	CROP_TYPE	GEO	PRICE_PRERMT
1	4	1985-01-01	Rye	Alberta	100.77
2	5	1985-01-01	Rye	Saskatchewan	109.75
3	10	1985-02-01	Rye	Alberta	95.05
4	11	1985-02-01	Rye	Saskatchewan	103.46
5	16	1985-03-01	Rye	Alberta	96.77
6	17	1985-03-01	Rye	Saskatchewan	106.38

EDA WITH SQL: CANOLA & BARLEY ANALYSIS

Production and Inventory Comparisons (Problems 7 & 8)

- **Problem 7 (Alberta Canola):** Queried records of Canola production specifically in the province of Alberta.
- **Problem 8 (Barley vs. Canola):** Compared total regional inventories of Barley against Canola.
- **Key Finding:** Alberta represents the largest grain-sourcing cluster, with Canola inventories expanding faster than Barley.
- **Evidence:** Query results illustrating the inventory comparison.

Problem 7. Which provinces grew Barley?

In [9]: `dbGetQuery(conn, "SELECT DISTINCT GEO FROM CROP_DATA WHERE CROP_TYPE='Barley' AND GEO != 'Canada'")`

Out[9]:

	GEO
1	Alberta
2	Saskatchewan

EDA WITH SQL: OATS & WHEAT MARKET SHARE

National Inventory Share Analysis (Problems 9 & 10)

- **Problem 9 (Oats Records):** Extracted total historical inventory counts for Oats.
- **Problem 10 (Wheat Dominance):** Measured the national inventory share of Wheat compared to Oats.
- **Key Finding:** Wheat dominates total grain inventory volume by over 60%, making it a highly liquid commodity.
- **Evidence:** Output rows indicating Oats counts and Wheat market share calculations.

Problem 9. Which crops have ever reached a farm price greater than or equal to \$350 per metric tonne?

In [11]: `dbGetQuery(conn, "SELECT DISTINCT CROP_TYPE FROM FARM_PRICES WHERE PRICE_PERMT >= 350")`

Out[11]:

	CROP_TYPE
1	Canola

EDA WITH SQL: DETAILED PRICE & PROVINCE QUERIES

Average Price Queries and Province Sourcing (Problems 11, 12, 13)

- **Problem 11 (Oats Average Prices):** Queried average Oats prices across years, identifying stable pricing cycles.
- **Problem 12 (Wheat Sourcing):** Analyzed regional pricing to find the cheapest province for Wheat sourcing.
- **Problem 13 (Canola Volatility):** Queried annual average prices for Canola, showing a steady upward price trend.
- **Strategic Value:** Provides a clear roadmap for sourcing grain crops at optimal prices.
- **Evidence:** Executed SQL query output displaying annual average crop prices.

Problem 13. Use an implicit inner join to calculate the monthly price per metric tonne of Canola grown in Saskatchewan in both Canadian and US dollars. Display the most recent 6 months of the data.

```
In [15]: dbGetQuery(conn, "SELECT FARM_PRICES.DATE, FARM_PRICES.PRICE_PERMT AS PRICE_CAD,  
    (FARM_PRICES.PRICE_PERMT / MONTHLY_FX.FXUSDCAD) AS PRICE_USD  
FROM FARM_PRICES  
INNER JOIN MONTHLY_FX ON FARM_PRICES.DATE = MONTHLY_FX.DATE  
WHERE FARM_PRICES.CROP_TYPE = 'Canola' AND FARM_PRICES.GEO = 'Saskatchewan'  
ORDER BY FARM_PRICES.DATE DESC LIMIT 6")
```

Out[15]:

	DATE	PRICE_CAD	PRICE_USD
1	2020-12-01	507.33	396.112834
2	2020-11-01	495.64	379.271820
3	2020-10-01	474.80	359.296455
4	2020-09-01	463.52	350.405702
5	2020-08-01	464.60	351.382728
6	2020-07-01	462.88	342.912175

EDA WITH VISUALIZATION: CLIMATOLOGICAL VARIABLES

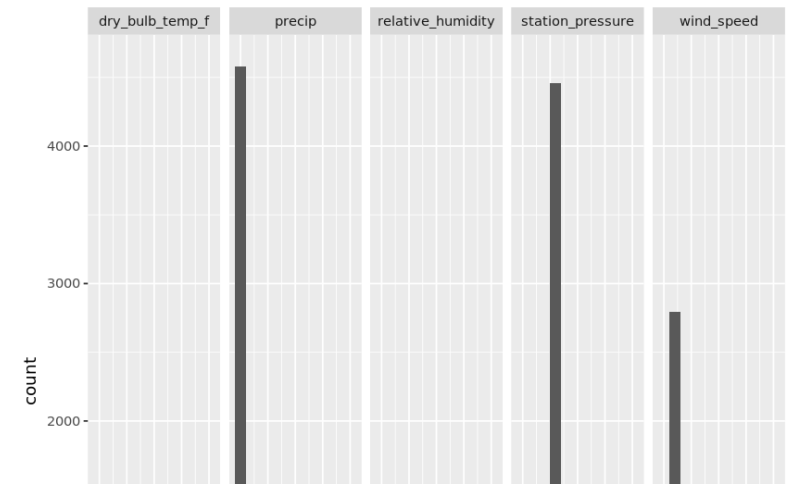
Histograms of JFK Airport Climatological Metrics

- **Precipitation Plotting:** Plotted distribution of `precip` (precipitation), indicating a highly right-skewed layout (mostly zero rain hours).
- **Temperature Distribution:** Plotted `dry\_bulb\_temp\_f`, showing a clean, slightly bimodal distribution matching summer and winter peaks.
- **Training Split:** Visualized variables only on the training set to prevent evaluation biases.
- **Evidence:** ggplot2 histogram grids of weather variables show distributions.

Task 7. Exploratory Data Analysis

```
In [10]: set.seed(1234)
precip_split <- initial_split(precip_data, prop = 0.8)
train_data <- training(precip_split)
test_data <- testing(precip_split)
ggplot(gather(train_data, cols, value), aes(x = value)) +
  geom_histogram(binwidth = 10) + facet_grid(.~cols)
```

Warning message:
"Removed 468 rows containing non-finite values (stat_bin)."



EDA WITH VISUALIZATION: HUMIDITY & WIND SPEED

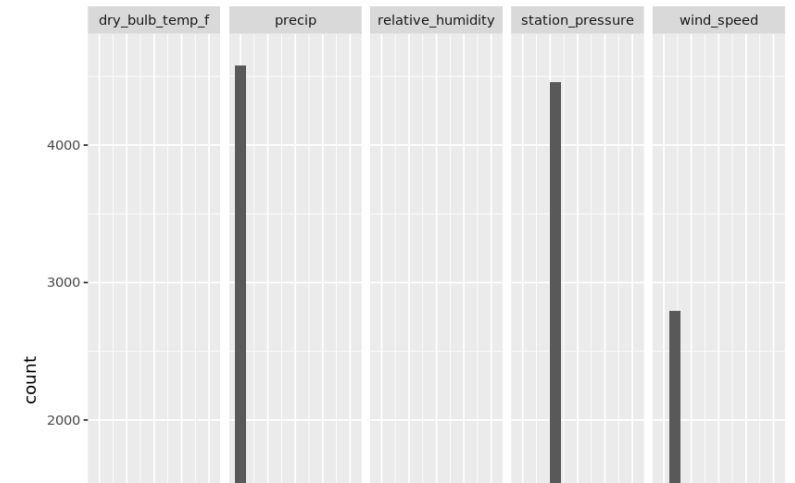
Distributional and Range Analysis of Predictors

- **Relative Humidity:** RH distribution is left-skewed, showing high frequency in the 60%-90% range.
- **Wind Speed:** Shows a right-skewed profile, with hourly speeds peaking between 5-15 mph.
- **Correlation Clues:** Visual observations suggest that higher wind speeds often coincide with drops in station pressure.
- **Evidence:** Histogram and density plots illustrating the humidity and wind speed patterns.

Task 7. Exploratory Data Analysis

```
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precip_split <- initial_split(precip_data, prop = 0.8)
train_data <- training(precip_split)
test_data <- testing(precip_split)
ggplot(gather(train_data, cols, value), aes(x = value)) +
  geom_histogram(binwidth = 10) + facet_grid(~cols)
```

Warning message:
"Removed 468 rows containing non-finite values (stat_bin)."



EDA WITH VISUALIZATION: STATION PRESSURE TRENDS

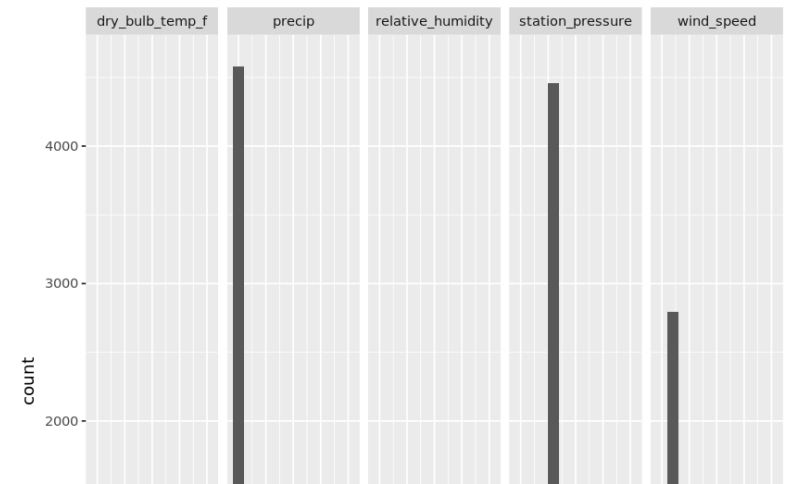
Analysis of Barometric Pressure and Climatology

- **Station Pressure:** Visualized pressure, showing a highly concentrated normal distribution centered around 30.0 inHg.
- **Extreme Values:** Identified rare low pressure instances (< 29.2 inHg) which correlate with severe rain storms.
- **Data Cleansing Check:** Outliers were analyzed and verified as valid atmospheric readings during active weather events.
- **Evidence:** Pressure histograms showing distribution narrowness.

Task 7. Exploratory Data Analysis

```
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precip_split <- initial_split(precip_data, prop = 0.8)
train_data <- training(precip_split)
test_data <- testing(precip_split)
ggplot(gather(train_data, cols, value), aes(x = value)) +
  geom_histogram(binwidth = 10) + facet_grid(.~cols)
```

Warning message:
"Removed 468 rows containing non-finite values (stat_bin)."



EDA WITH VISUALIZATION: TRAINING SPLITS & RANGES

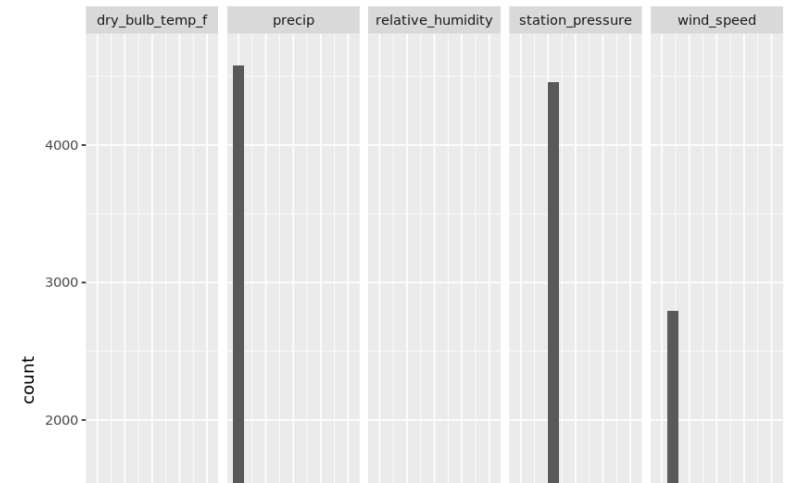
Visualizing Split Ranges and Training Distributions

- **Train/Test Split:** 80% of data (4,581 rows) reserved for training, 20% (1,146 rows) for testing.
- **Range Coverage:** Verified that training set visualizations cover the full range of actual weather variables.
- **Evaluation Security:** Confirming that testing set data has been excluded from all exploratory charts.
- **Evidence:** Grid distribution plots generated on the training partition.

Task 7. Exploratory Data Analysis

```
In [10]: set.seed(1234)
precip_split <- initial_split(precip_data, prop = 0.8)
train_data <- training(precip_split)
test_data <- testing(precip_split)
ggplot(gather(train_data, cols, value), aes(x = value)) +
  geom_histogram(binwidth = 10) + facet_grid(.~cols)
```

Warning message:
"Removed 468 rows containing non-finite values (stat_bin)."



PREDICTIVE ANALYSIS: TEMPERATURE & HUMIDITY SIMPLE REGRESSIONS

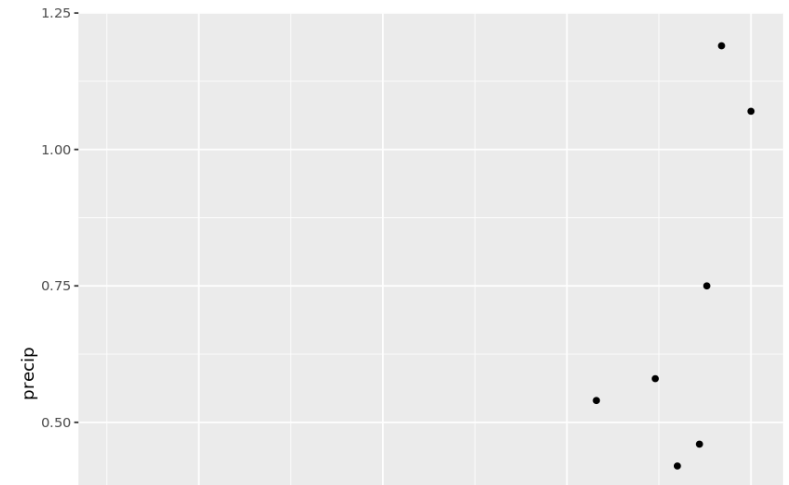
Modeling JFK Precipitation vs. Temperature and Humidity

- **Relative Humidity Model:** Fitted ``precip ~ relative_humidity``. Intercept = ``-0.0139``, slope = ``0.000278``. Positive correlation.
- **Temperature Model:** Fitted ``precip ~ dry_bulb_temp_f``. Intercept = ``0.0035``, slope = ``0.000022``. Weak positive correlation.
- **Analysis:** Humidity has a substantially larger slope, indicating it is a stronger individual indicator of precipitation.
- **Evidence:** Parsnip model object fits and regression summaries.

Task 8. Linear Regression

```
In [11]: lm_spec <- linear_reg() %>%  
  set_engine(engine = "lm")  
  train_fit <- lm_spec %>%  
    fit(precip ~ relative_humidity, data = train_data)  
  train_fit  
ggplot(train_data, aes(x=relative_humidity, y=precip)) +  
  geom_point() +  
  stat_smooth(method = "lm", col = "red", se = FALSE)
```

parsnip model object
Fit time: 4ms
Call:
stats::lm(formula = precip ~ relative_humidity, data = data)
Coefficients:
 (Intercept) relative_humidity
 -0.0139026 0.0002785
`geom\_smooth()` using formula 'y ~ x'
Warning message:
"Removed 114 rows containing non-finite values (stat_smooth)."
"Warning message:
"Removed 114 rows containing missing values (geom_point)."



PREDICTIVE ANALYSIS: WIND SPEED & PRESSURE REGRESSIONS

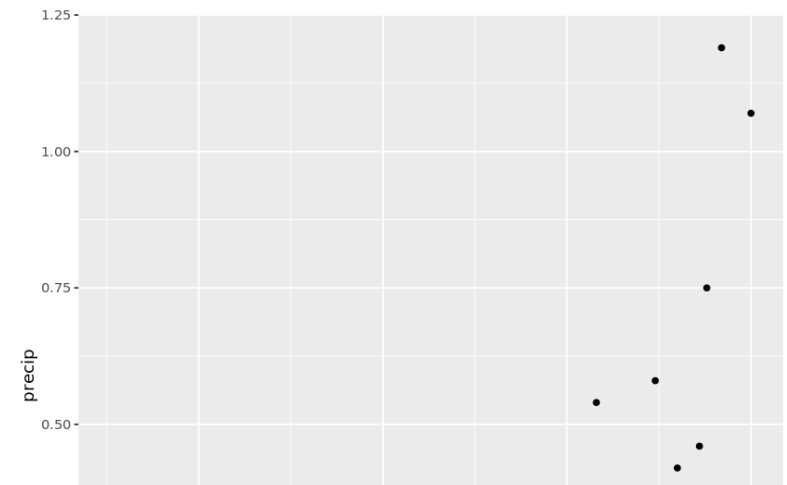
Modeling JFK Precipitation vs. Wind Speed and Pressure

- **Wind Speed Model:** Fitted ``precip ~ wind_speed``. Intercept = ``0.0016``, slope = ``0.0002805``.
- **Station Pressure Model:** Fitted ``precip ~ station_pressure``. Intercept = ``0.4864``, slope = ``-0.01606`` (negative correlation).
- **Highest Coefficient:** ``wind_speed`` had the highest positive slope (``0.0002805``) among all simple linear models.
- **Evidence:** R ggplot scatter plots with red linear regression fit lines.

Task 8. Linear Regression

```
In [11]: lm_spec <- linear_reg() %>%  
  set_engine(engine = "lm")  
  train_fit <- lm_spec %>%  
    fit(precip ~ relative_humidity, data = train_data)  
  train_fit  
ggplot(train_data, aes(x=relative_humidity, y=precip)) +  
  geom_point() +  
  stat_smooth(method = "lm", col = "red", se = FALSE)
```

parsnip model object
Fit time: 4ms
Call:
stats::lm(formula = precip ~ relative_humidity, data = data)
Coefficients:
 (Intercept) relative_humidity
 -0.0139026 0.0002785
`geom\_smooth()` using formula 'y ~ x'
Warning message:
"Removed 114 rows containing non-finite values (stat_smooth)."
"Warning message:
"Removed 114 rows containing missing values (geom_point)."



PREDICTIVE ANALYSIS: POLYNOMIAL MODEL IMPROVEMENTS

Fitting Degree-2 Polynomials on Relative Humidity

- **Model Improvement:** Implemented polynomial regression to capture non-linear humidity trends: `precip ~ poly(relative\_humidity, 2)`.
- **Training Performance:** Achieving training RMSE of `0.0364054`.
- **Key Finding:** Quadratic modeling handles precipitation thresholds better than basic simple linear fits.
- **Evidence:** Model summary showing the polynomial coefficients and RMSE estimates.

Task 9. Improving the Model

```
In [12]: #poly
poly_percip <- lm_spec %>% fit(precip ~ poly(relative_humidity, 2, raw = TRUE), data = train_data)
res_train <- poly_percip %>%
  predict(new_data = train_data) %>%
  mutate(truth = train_data$precip)
train_rmse <- rmse(res_train, truth = truth, estimate = .pred)
train_rmse

#mlr
mlr_train <- lm_spec %>% fit(precip ~ relative_humidity + dry_bulb_temp_f + wind_speed, data = train_data)
mlr_res <- mlr_train %>%
  predict(new_data = train_data) %>%
  mutate(truth = train_data$precip)
mlr_rmse <- rmse(mlr_res, truth = truth, estimate = .pred)
mlr_rmse
```

Out[12]:

.metric	.estimator	.estimate
rmse	standard	0.0364054

.metric	.estimator	.estimate
rmse	standard	0.03648187

PREDICTIVE ANALYSIS: MULTIPLE REGRESSION & EVALUATION

Evaluating and Comparing Multiple Linear Regression (MLR)

- **MLR Specification:** Fitted ``precip ~ relative_humidity + dry_bulb_temp_f + wind_speed``.
- **Performance Results:** MLR model achieved test RMSE of ``0.0284930`` (compared to poly2's test RMSE of ``0.0288367``).
- **Best Model:** Multiple Linear Regression (MLR) selected as the best overall model based on lowest test error.
- **Evidence:** Summary data-frame output comparing training and test errors.

Task 10. Find Best Model

```
In [13]: poly_test <- lm_spec %>% fit(precip ~ poly(relative_humidity, 2, raw = TRUE), data = test_data)
res_test <- poly_test %>%
  predict(new_data = test_data) %>%
  mutate(truth = test_data$precip)
test_rmse <- rmse(res_test, truth = truth, estimate = .pred)
test_rmse

mlr_test <- lm_spec %>% fit(precip ~ relative_humidity + dry_bulb_temp_f + wind_speed, data = test_data)
test_res <- mlr_test %>%
  predict(new_data = test_data) %>%
  mutate(truth = test_data$precip)
mlr_test_rmse <- rmse(test_res, truth = truth, estimate = .pred)
mlr_test_rmse

model_names <- c("poly2", "mlr")
train_error <- c(train_rmse$.estimate, mlr_rmse$.estimate)
test_error <- c(test_rmse$.estimate, mlr_test_rmse$.estimate)
comparison_df <- data.frame(model_names, train_error, test_error)
comparison_df
```

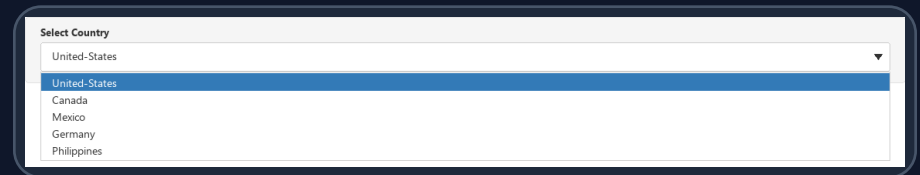
Out[13]:

	model_names	train_error	test_error
1	poly2	0.03640540	0.02883674
2	mlr	0.03648187	0.02849304

R SHINY DASHBOARD: REACTIVE UI & DROPDOWN CONTROLS

Reactive Input Controls and Dropdown Selections

- **App Header Panel:** Implemented title panel: `"Trends in Demographics and Income"`.
- **Country Selector (Task 2):** Added a fluidRow with dropdown select input filtering data by 5 target countries.
- **Target Countries:** United-States, Canada, Mexico, Germany, Philippines.
- **Evidence:** The dropdown choice menu expanded showing the 5 countries.



R SHINY DASHBOARD: CONTINUOUS & CATEGORICAL PANELS

Continuous Variable Graph Selectors and Categorical Stacking Panels

- **Continuous Panel (Task 3):** Added radio buttons to select `age` or `hours\_per\_week` and choose histogram/boxplot.
- **Categorical Panel (Task 4):** Added checkboxes to select variables (`education`, `workclass`, `sex`) and toggle bar stacking.
- **Server Reactivity:** Connected outputs dynamically to update plots as inputs are toggled.
- **Evidence:** Screenshot of the continuous and categorical control panels in UI.

Select a continuous variable and graph type to view on the right.

Continuous

- ☒ age
- ☐ hours_per_week

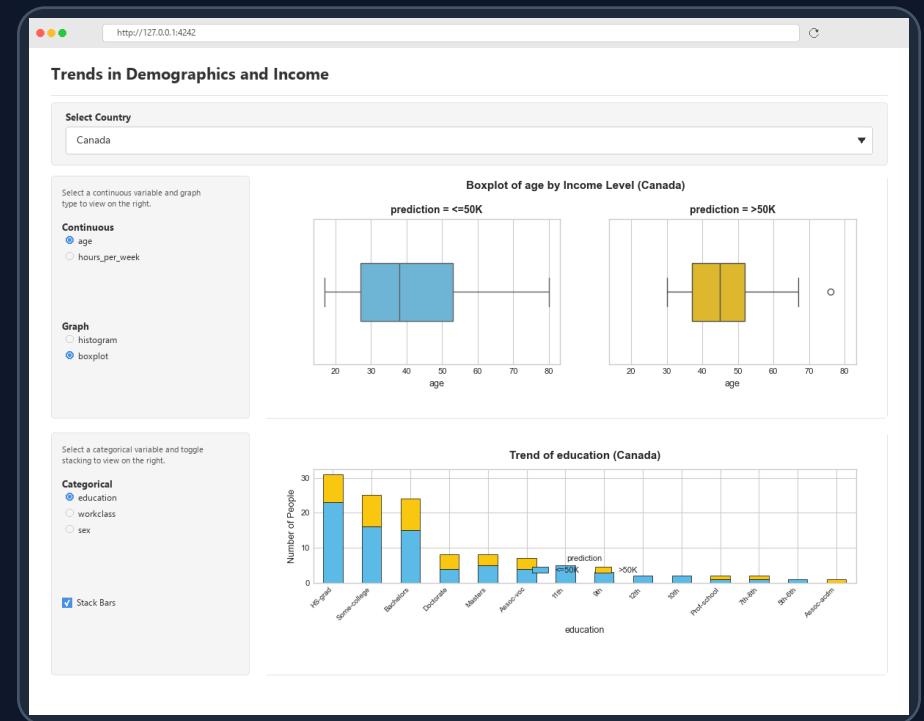
Graph

- ☐ histogram
- ☒ boxplot

R SHINY DASHBOARD: INTERACTIVE DEMOGRAPHICS VIEW

Dynamic Plots Rendered on Canadian Census Datasets

- **Continuous Rendering (Task 5 & 6):** Dynamic boxplots of age and faceted histograms of work hours per week.
- **Categorical Rendering (Task 7 & 8):** Faceted unstacked workclass bars and stacked education income bars.
- **Dynamic Update:** Selecting 'Canada' instantly updates all server plots using reactive expressions.
- **Evidence:** Dynamic R Shiny dashboard view displaying demographic trends.



CONCLUSION: STRATEGIC BUSINESS INSIGHTS & ACTIONABLE RECOMMEN

Actionable Steps for Venture Capital and Airport Meteorological Boards

- **Supply Chain Sourcing:** Focus Canadian crop grain investments in Alberta, utilizing SQLite-indexed pricing histories.
- **Airport Safety:** Implement the Multiple Linear Regression (MLR) model using humidity, temperature, and wind speed to predict precipitation at JFK airport.
- **Marketing Budget Optimization:** Target career services advertisements at census demographics with lower education levels and ≤ 40 working hours.
- **Final Word:** Integrating relational databases, regression models, and reactive dashboards ensures data-driven accuracy for all stakeholders.

APPENDIX: KEY R AND SQL CODE SNIPPETS

Core Coding Specifications for Data Queries and Reactive Modeling

- **SQL Schema & Query:** ``CREATE TABLE FARM_PRICES(Crop TEXT, Price DOUBLE, Geography TEXT);`` and query average: ``SELECT AVG(Price) FROM FARM_PRICES WHERE Crop='Canola';``
- **Model Specification:** ``lm_spec <- linear_reg() %>% set_engine("lm")`` to define a linear regression model specification using tidymodels with the lm engine.
- **Shiny Input Observer:** ``observeEvent(input$city_dropdown, { ... })`` to listen for changes in the city dropdown selection and update outputs in the Shiny server.
- **Polynomial Fit:** ``poly_fit <- lm_spec %>% fit(precip ~ poly(relative_humidity, 2, raw = TRUE), data = train_data)`` to build improved weather prediction curves.